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Dual Purpose Parachute Recovery System, rev2

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Introduction

One of the more challenging ESRA requirements is to recover intact all flight hardware. We do not want to litter BLM land with rocket junk. Based on considerable flight operations experience, this is nowhere as easy as it sounds. Most ESRA parachute recovery systems first deploy a drogue parachute to reduce the dynamic pressure to a level which is safe for main chute deployment. Unfortunately, this is not always a practical approach. This memo addresses a new recovery scheme suggested by Adam Vore, our parachute expert.

The Vore scheme separates the rocket body into two parts just below the desired apogee. The two body parts are a heavy solid nose section and a lighter aft body containing all the electronics. At separation the parachute, attached to the light aft body, is deployed for apogee control. The two body halves are attached via a lanyard, including a length of bungee cord. The impact shock to the aft body is mitigated because the aft body will slow down significantly after the heavy nose has impacted.

This memo documents the analysis for the descent time, aft body impact velocity and lanyard length. A parachute reefing option is described.

Nomenclature

Symbol	Definition
V	Descent velocity, ft/s
h	Altitude AGL, of the aft piece, ft
t	Time from parachute deployment, s
m_1	Mass of the solid nose, sl
m_2	Mass of the aft body, including the parachute, sl
C_D	Drag coefficient of the disreefed parachute,
C_{DR}	Drag coefficient of the reefed parachute,
S	Parachute projected area, ft^2,
g	Acceleration cue to gravity, ft/s^2,
ρ	Mass density of the atmosphere, sl/ft^3,
α	Atmospheric temperature lapse rate = – 0.003566 $^oR/ft$,
R	Gas constant for air = 1716 $ft^2 / s^2\ ^oR$
T	Atmospheric temperature, ^oR
L	Tether length, ft
τ	Reefing ratio = Diameter of reefing line circle / Nominal parachute diameter
$( )_o$	Conditions at sea level
$( )_D$	Conditions at parachute deployment
$(.)_{DR}$	Conditions at parachute disreefing

$( )_I$ Conditions at impact of the nose piece

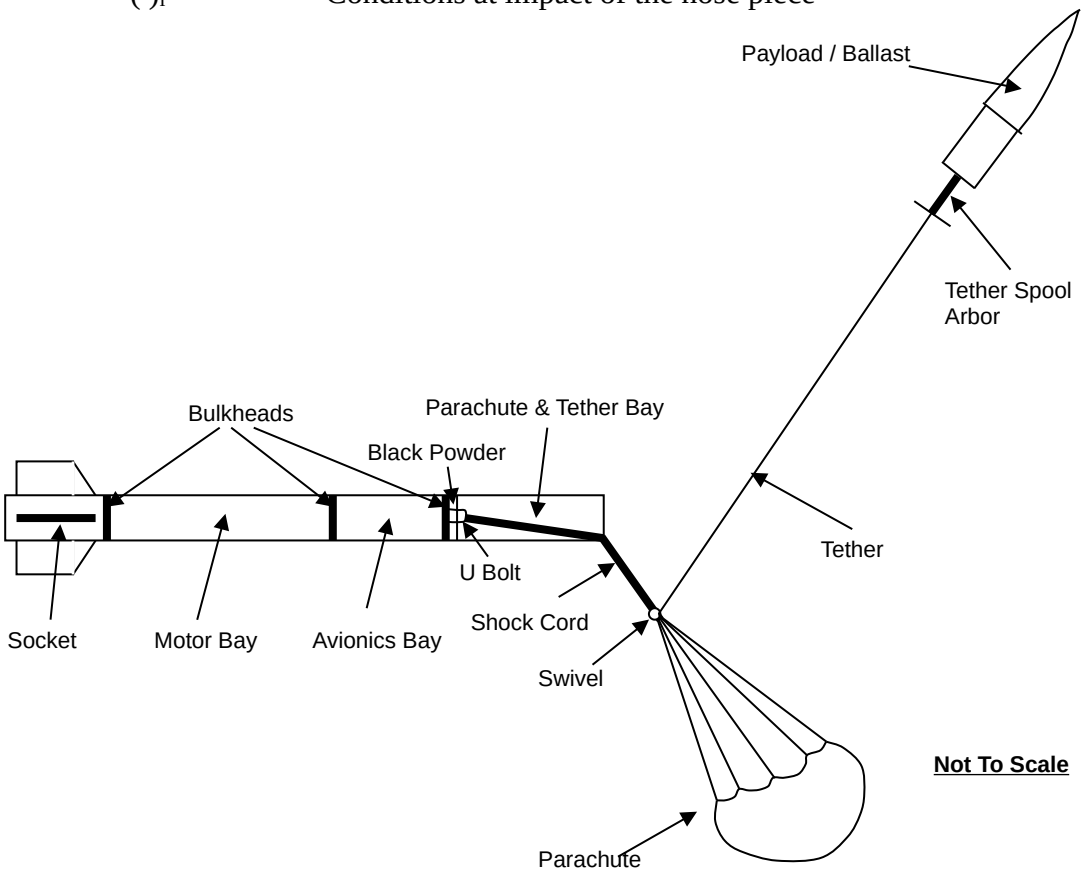


Figure 1 Second Stage with Deployed Parachute

Operations Description

Let's start with a description of the second stage, and how it works. A graphical depiction of Stage 2 at parachute deployment is shown in Fig. 1. At its extreme aft end is the socket bay that receives the interstage spigot, part of the first stage. Just forward of this space is the bay housing the second stage H motor along with its ignition wiring and guide tube. The avionics bay is forward of the motor bay, and contains the Arduino flight computer, the GPS radio homing beacon and sensors. These bays are bounded by birch plywood bulkheads, the aftermost having a circular cutout for the rocket motor.

The forward part of the second stage is devoted to recovery. A single parachute is used, initially deployed in a reefed condition. The parachute is held in place by the payload / ballast element which is attached to the second stage hull by nylon shear screws. Upon computer command, a small powder charge behind the parachute is ignited. The gases push the parachute bag which, in turn, pushes the payload / ballast shearing the nylon retention screws. As the payload / ballast moves forward, a tether coiled around its arbor deploys. The other end of the tether is attached to the parachute swivel. When the tether is fully deployed, it extracts the parachute. Finally, the parachute swivel is connected to the second stage hull via a shock cord. Note the care must be taken to preclude abrasion damage to the shock cord where it rubs against the open mouth of the second stage. The parachute is disreefed at a modest altitude above

the ground and the assembly slows to a new equilibrium descent speed. Finally, during landing the payload / ballast first contacts the ground causing the tension in the tether to vanish. When this happens the second stage hull is slowed to its final impact speed.

Tether Spool

The tether material is assumed to be 1/8" nylon parachute cord wrapped in layers around a central arbor. Consider a 1/8" length of spool in which the tether is wrapped around an arbor of 1/4" diameter. The following table summarizes the tether storage:

Tether layer	Mean layer radius, in	Layer length, in
1	0.1875	1.178
2	0.3125	1.963
3	0.4375	2.749
4	0.5625	3.534
5	0.6875	4.320
6	0.8125	5.105
7	0.9375	5.890
Total		27.740

Table 1 Tether Storage

Thus, a spool 1/8" long can store about 2 ' of tether. A 2" long spool carries 33 ft of tether.

Shock Cord

Attach the second stage hull U bolt to the parachute swivel with a length of shock cord to mitigate opening shock loads. Some thoughts on how this works are in order: First there are two possible scenarios for parachute deployment.

In the first, the black powder applies a pressure force to the aft end of the parachute, and it in turn shears the nylon screws so that the payload / ballast comes out first followed by the folded parachute. If the payload / ballast separates at modest velocity, say, up to 10 ft/sec, then the parachute will be fully inflated before the 33 ft tether is fully deployed.. Assuming deployment on the ascending leg, the parachute will first strongly pull the rocket body downward, and then, after the tether is deployed the rocket body & payload / ballast will be strongly pulled upward. The magnitude of the first downward pull can be estimated using OSCALC. The magnitude of the second, upward pull could be estimated given the payload / ballast separation velocity (currently unknown).

In the second scenario, the payload / ballast is separated as in the first scenario, but the parachute hangs up inside the rocket body. When the tether is fully extended (33 ft) the payload / ballast will jerk the parachute the rest of the way out. This is essentially a backup deployment scenario. In this case, the opening shock given by OSCALC will be divided between pulling the rocket body and the payload / ballast down. It might be a more severe load on the tether.

Analysis of Descent

During steady state descent Newton's third Law is

$$(m_1 + m_2)g = (1/2)C_{DR}\rho S V^2 \text{, or}$$

$$V = \frac{dh}{dt} = -\sqrt{\frac{2(m_1 + m_2)g}{C_{DR}\rho S}}\,. \tag{1}$$

Given density as a function of altitude, this can be integrated to find the descent time

$$\rho = \rho_o\left(1 + \frac{\alpha h}{T_o}\right)^{-\left(\frac{1}{\alpha R}\right)-1} \text{ in the troposphere.} \tag{2}$$

Then,
$$\left(1 + \frac{\alpha h}{T_o}\right)^{\left(\frac{1}{2\alpha R}\right)+\frac{1}{2}} dh = \sqrt{\frac{2(m_1 + m_2)g}{C_{DR}\rho_o S}}\,dt \text{, and}$$

$$\left(1 + \frac{\alpha h_{DR}}{T_o}\right)^{\frac{1+3\alpha R}{2\alpha R}} - \left(1 + \frac{\alpha h_D}{T_o}\right)^{\frac{1+3\alpha R}{2\alpha R}} = \left(\frac{1+3\alpha R}{2RT_o}\right)\sqrt{\frac{2(m_1 + m_2)g}{C_{DR}\rho_o S}}\,\Delta t\,. \tag{3}$$

The descent time, Δt ,is important because the larger this becomes the greater the wind drift during descent. If the parachute is not reefed,

Δt = Total time from deployment to disreefing $= t_{DR} - t_D$

Otherwise, the total time from deployment to impact is the time from deployment to disreef plus the time from disreef to impact. Note that the parachute drag area will change as a result of disreefing. We use eq. (3) to find these two separate times.

Total time $= (t_{DR} - t_D) + (t_I - t_{DR})$ (4)

Analysis of Impact

As soon as the heavy solid nose has stopped, the aft piece and parachute slow down. Newton's Second Law for this situation is:

$$m_2 \frac{dV}{dt} = m_2 g - (1/2)C_D \rho S V^2 \tag{5}$$

Change the independent variable from time to altitude:

$$-(1/2)m_2\frac{dV^2}{dh}=m_2g-(1/2)C_D\rho SV^2$$

(6)

The relevant initial condition is

$$V^2=V_I^2 \text{ at } h=h_I+L$$

This can be easily integrated to give

$$V^2=\frac{2m_2g}{C_D\rho S}+\frac{2m_1g}{C_D\rho S}\exp(\frac{C_D\rho S}{m_2}(-h_I-L+h))$$

When the aft piece strikes the ground, its velocity will be

$$V^2=\frac{2m_2g}{C_D\rho S}+\frac{2m_1g}{C_D\rho S}\exp(-\frac{C_D\rho SL}{m_2}) \quad (7)$$

This model has been incorporated into ONECHUTE3.xls.

Parachute

Now suppose the following values apply:

Payload / ballast weight	10 lbs
Aft body weight	8.5 lbs
Tether length	33 ft
Deployment altitude	25,000 ft AGL
Disreef altitude	1,000 ft AGL
Impact altitude	4300 ft MSL
Parachute description	LOC Precision Classic II size 44 (5.18 ft dia)
Reefing ratio	0.25

Table 2 Second Stage Recovery data

The selection of reefing ratio merits some discussion. Larger reefing ratios imply smaller drag and faster descent times (good to minimize wind drift) However, they also imply higher speeds at disreef (Greater risk to the parachute). The values selected above result in:

Sea level descent speed (reefed)	61.25 ft/sec
Total descent time	27.1 minutes
Aft body impact speed	20.86 ft/sec

Table 3 Second stage recovery performance

Summary

This study describes a second stage recovery concept and identifies typical parameters. The major outstanding risk is parachute deployment.

References

1."Reefing of Parachutes", by T.W. Knacke, Aeronautical Systems Division report AD-A036 497, 1976

2. "Rocket Recovery Redefined", anon, LOC Precision data sheets, 2013

3. “OSCALC – Opening Shock Calculator, Version 1.01, User’s Manual”, G. Peek and J. Potvin, Parks College Parachute Research Group, Saint Louis University, St. Louis, MO